

Supervised Pre-trained Model Fine-tuning Chinese Corpus for Gender Bias Detection, Classification and Mitigation Challenge

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Abstract. As artificial intelligence and natural language processing (NLP) systems become increasingly integrated into our daily lives, addressing gender bias in these systems has become crucial for ensuring fair and equitable technology. Our team mainly participated in Subtask 1: Bias Detection, requiring the model to judge whether a sentence is gender-biased (B), or non-biased (N). and Subtask 2: Bias Classification, which involves classifying the gender-biased sentences into different categories, including gender stereotyped activity and career choices (AC), gender stereotyped descriptions and inductions (DI), and expressed gender stereotyped attitudes, norms and beliefs (ANB). We mainly used various fine-tuning pre-trained models or ensemble learning methods of pre-trained models and achieved remarkable results in this field, ranking 4th and 3rd respectively in the two subtasks. Corresponding to the task setting team and the official dataset results from NLPCC 2025 Shared Task 7-CORGI-PM 2025: Chinese Corpus for Gender Bias Detection, Classification and Mitigation Challenge. The project source code is available from https://github.com/wangkongqiang/nlpcc2025_task7.

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1. Introduction

NLPCC 2025¹ will continue the tradition of previous NLPCC conferences by hosting several shared tasks in natural language processing (NLP) and Chinese computing. This year, the shared tasks will focus on both classic problems and newly emerging challenges. These include LLM-Generated Text Detection, Evaluation of On-Topic Graded Comments for Essays, Comprehensive Argument Analysis for Chinese Argumentative Essays, Multi-modal, Multilingual, and Multi-hop Medical Instructional Video Question Answering Challenge, Chinese Government Text Correction using Knowledge Bases, Faithful Bilingual Long-context Retrieval and Generation Challenge, Chinese Corpus for Gender Bias Detection, Classification, and Mitigation Challenge, and Personalized Emotional Support Conversation.

Based on these task descriptions and the official datasets, we conducted a cross-temporal study on the content of Chinese language corpora, and achieved remarkable success in Chinese Corpus for Gender Bias Detection, Classification, and Mitigation Challenge.

1.1. Task Introduction

There are eight shared tasks in this year's NLPCC conference and the detailed description of each task can be found in the task guidelines released later. Our team mainly participates in Task 7-CORGI-PM 2025: Chinese Corpus for Gender Bias Detection, Classification and Mitigation Challenge [1]. As artificial intelligence (AI) and natural language processing (NLP) systems become increasingly integrated into our daily lives, addressing gender bias in these systems has become crucial for ensuring fair and equitable technology. Following the groundbreaking work of CORGI-PM, which established the first sentence-level chinese corpus for gender bias probing and mitigation, they propose a comprehensive shared task to advance the state-of-the-art in automatic gender bias detection, classification, and mitigation in chinese text. The challenge consists of three tracks that progressively build upon each other. The specific details of these tracks are as follows:

- Track 1: Gender Bias Detection. Binary classification task to identify whether a given chinese text contains gender bias. Evaluation metrics: Precision, Recall, F1-score.
- Track 2: Gender Bias Classification. Multi-label classification of identified gender bias into three categories: Gender Stereotyped Activity and Career Choices (AC), Gender Stereotyped Descriptions and Inductions (DI), Expressed Gender-stereotyped Attitudes,

¹ <http://tcci.ccf.org.cn/conference/2025/index.php>

Norms and Beliefs (ANB). Evaluation metrics: Class-wise and macro-averaged Precision, Recall, F1-score.

- Track 3: Gender Bias Mitigation. Natural language generation task to rewrite biased text while preserving core meaning, Evaluation using BLEU, METEOR, and ROUGE-L.

Training set includes enhanced version of CORGI-PM with 30,000+ annotated sentences, development set includes 1,500 sentences, hidden test set includes 1,500 sentences. All data manually annotated by qualified annotators with relevant educational backgrounds. This to some extent ensures the successful execution of our special task experiments and the validity of the data sets we used.

1.2. Related Work

1.2.1. Sentence Embedding

When chinese corpus for gender bias detection and classification is needed, we are directed to use the ready-made tool frameworks that need to be considered. Word embedding models such as BERT [2], GloVe [3], RoBERTa [4], and Word2Vec [5] are frequently employed to assess the semantic distance between words. They are also some of the more commonly used methods in text classification tasks in a specific field. Sentence embeddings with a fixed length are often generated via mean/max pooling of word embeddings or employing CLS embedding in BERT. The semantic distances are commonly measured using the cosine similarity of embeddings of two expressions. Siamese or triplet network architectures are frequently employed in sentence embedding training. For example, models such as Sentence-BERT utilize a dual-encoder architecture with shared weights for predicting sentence relationships e.g., semantic contradiction, entailment, or neutral labeling and for similarity score prediction using regression objectives, e.g., the difference between human annotated similarity score (sim) of two sentences and the cosine of two sentence embeddings [6].

1.2.2. Ensemble Learning

In previous studies [7], ensemble learning presents several advantages. The ensemble approach can reduce the errors from individual models by combining results from multiple sources or can make the system more robust. In our study, using multiple pre-trained models can also save a substantial amount of computation while making use of information from the large amount of open access dataset during pre-training. Previous research [8] has demonstrated that ensemble learning can achieve remarkable success in many text detection and classification tasks.

In our study, we aim to integrate multiple pre-trained models fine-tuning learning to assess semantic relatedness. When models are trained on diverse datasets with different architectures, they may produce varied predictions on semantic relatedness, and combining them may improve overall performance. We use sentence embeddings mainly from the following pre-trained models. Multilingual BERT (cased, uncased), RoBERTa, BETO (cased, uncased), DistilBERT and so on. Tables 1, 2, and 3 give a summary of all pre-trained models in subtask 1 and subtask 2 respectively.

Table 1. Pre-trained models are used in Subtask 1: Gender Bias Detection.

Pre-trained Model Name	Model Learning Situation	Model Learning Name
models--clue--roberta-chinese-large	Fine-tuning Learning	roberta-chinese-large
models--Geotrend--bert-base-zh-cased	Integrated Learning	Bagging
models--nghuyong--ernie-1.0-base-zh	Integrated Learning	Bagging
models--hongyin--roberta-chinese-homie-large	Integrated Learning	Bagging
models--clue--roberta-chinese-large	Integrated Learning	Bagging
models--Geotrend--bert-base-zh-cased	Integrated Learning	Bagging2
models--hongyin--roberta-chinese-homie-large	Integrated Learning	Bagging2
models--clue--roberta-chinese-large	Integrated Learning	Bagging2

Table 2. Pre-trained models are used in Subtask 2: Gender Bias Classification

(There are official test results).

Pre-trained Model Name	Model Learning Situation	Model Learning Name
models--distilbert--distilbert-base-multilingual-cased	Fine-tuning Learning	distilbert-multi
models--google-bert--bert-base-multilingual-cased	Integrated Learning	Bagging
models--dccuchile--bert-base-spanish-wwm-cased	Integrated Learning	Bagging
models--dccuchile--bert-base-spanish-wwm-uncased	Integrated Learning	Bagging
models--distilbert--distilbert-base-multilingual-cased	Integrated Learning	Bagging
models--google-bert--bert-base-multilingual-uncased	Integrated Learning	Bagging2
models--distilbert--distilbert-base-multilingual-cased	Integrated Learning	Bagging2

Table 3. Pre-trained models are used in Subtask 2: Gender Bias Classification

(There are no official test results).

Pre-trained Model Name	Model Learning Situation	Model Learning Name
models--Geotrend--bert-base-zh-cased	Fine-tuning Learning	bert-cased
models--Geotrend--bert-base-zh-cased	Integrated Learning	Bagging
models--nghuyong--ernie-1.0-base-zh	Integrated Learning	Bagging
models--clue--roberta-chinese-large	Integrated Learning	Bagging
models--Geotrend--bert-base-zh-cased	Integrated Learning	Bagging2
models--nghuyong--ernie-1.0-base-zh	Integrated Learning	Bagging2
models--hongyin--roberta-chinese-homie-large	Integrated Learning	Bagging2
models--clue--roberta-chinese-large	Integrated Learning	Bagging2

2. Methodology

2.1. Overall Architecture

2.1.1. Overall Architecture of Subtask 1 (Gender Bias Detection)

Bagging is a parallel ensemble learning method. Its main idea is to conduct retracted sampling from the original dataset, construct multiple base learners, and then combine the prediction results of these base learners through voting or averaging. The goal of bagging is to reduce variance, so it performs better on models such as decision trees and neural networks without pruning. Random Forest is a typical representative of bagging. It further enhances the generalization ability of the model by introducing random feature selection. The process of upgrading a weak learning algorithm to a strong learning algorithm involves repeatedly learning to obtain a series of weak classifiers such as decision trees and logistic regression, and combining these weak classifiers to obtain a strong classifier. The boosting algorithm involves two parts: the addition model and the forward stepwise algorithm. Additive model is a strong classifier formed by the linear addition of a series of weak classifiers. The general combination form is as follows:

$$F_M(x; P) = \sum_{m=1}^n \beta_m h(x; a_m) \quad (1)$$

Among them, $F_M(x; P)$ is the strong learner, a_m is the optimal parameter learned by the weak classifier, β_m is the proportion of the weak learner in the strong classifier, and P is all a_m and β_m the combination. These weak classifiers are linearly added together to form a

strong classifier. Forward stepwise algorithm is the classifier generated in the next iteration is trained on the basis of the previous round during the training process, namely:

$$F_m(x) = F_{m-1}(x) + \beta_m h(x; a_m) \quad (2)$$

2.1.2 Overall Architecture of Subtask 2 (Gender Bias Classification)

The pursued approach involves using a weighted voting system of ensembles composed of different transformers. We trained several state-of-the-art natural language processing (NLP) pre-trained models on a large amount of annotated text dataset to create ensembles of classifiers with different architectures and configurations. We then combined the predictions of these ensembles using a weighted voting system to produce the final predictions. We have used the following transformers for the ensembles: Multilingual BERT (cased, uncased), RoBERTa, BERT (cased, uncased), DistilBERT.

For each instance, the final classification decision is based on the weighted sum of these base transformer models outputs. The novel weighted-voting system presented involves using each normalized transformer's metric score in the ensemble F1-score or RMSE depending on the specific task, to assess the importance of these base transformer models in the final outputs of the ensemble as opposed to the arithmetic mean typically used in conventional voting systems.

2.2. Implementation Step

2.2.1 Implementation Step of Subtask 1 (Gender Bias Detection)

The basic principle of Subtask 1 (Gender Bias Detection) is that among all ensemble learning methods, the most intuitive one is majority voting. Because its purpose is to output the more popular or most popular prediction among the predictions of basic learners. Majority voting is the simplest ensemble learning technique, which allows the predictions of multiple basic learners to be combined. Similar to the working mode of elections, this algorithm assumes that

each basic learner is a voter and each category is a competitor. In order to select the competitor as the winner, the algorithm will consider voting. There are mainly two methods for combining multiple predictions with voting: one is hard voting, and the other is soft voting. We introduce two methods here. Hard voting combines many predictions by assuming that the category with the most votes is the winner. In a simple case with two categories (*Class A*, *Class B*) and three basic learners (*Learner 1*, *Learner 2*, and *Learner 3*), if the target category has at least two votes, it will become the final output of the integration method, see Figure 1. Soft voting takes into account the possibility of predicting categories. To combine the

prediction results, soft voting calculates the average probability of each category and assumes that the winner is the category with the highest average probability. If all classifiers can estimate the class probability. That is, they all have a *predict_proba()* method in Scikit-learn [9], then the average probability of the class can be calculated, and the voting classifier takes the class with the highest probability as its own prediction. This is called soft voting, a specific example see Figure 2.

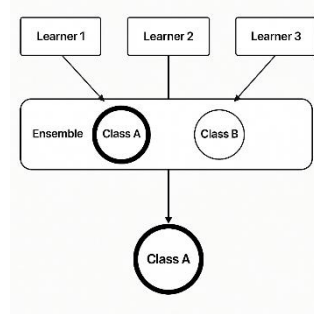


Fig. 1. Example of hard voting mechanism. Suppose *Class A* takes into account the votes from *Learner 1* and *Learner 2*, while *Class B* only accepts the vote from *Learner 3*. Therefore, *Class A* won by two votes over *Class B*.

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Model 1 A-99%; B-1%
Model 2 A-49%; B-51%
Model 3 A-40%; B-60%
Model 4 A-90%; B-10%
Model 5 A-30%; B-70%

A - (0.99 + 0.49 + 0.4 + 0.9 + 0.3) / 5
= 0.616

B - (0.01 + 0.51 + 0.6 + 0.1 + 0.7) / 5
= 0.384

Final result is A
  
```

Fig. 2. Example of soft voting mechanism. Suppose there are 5 models making predictions on the likelihood of *Class A* and *Class B*. The final average likelihood is obtained, with *Class A* having a likelihood of 0.616 which is greater than that of *Class B* (0.384). Therefore, *Class A* wins.

Hard Voting Classifier: The final result is determined based on the majority obeying the majority. **Soft Voting Classifier:** The average probability of all pre-trained models predicting samples to belong to a certain category is taken as the standard, and the type corresponding to the highest probability is the final prediction result. This kind of voting classifier often achieves a higher accuracy rate than a single best classifier. In fact, even if each classifier is a weak learner meaning it is only slightly better than random guessing, if there are enough weak learners and they are diverse enough, then the voting classifier ultimately integrated can still be a strong learner achieving high accuracy. In the support vector machine (SVM), the

parameter probability needs to be set to *True* to enable the support vector machine (SVM) to have the function of predicting class probabilities. In the voting classifier, voting needs to be set to *soft*. Generally, *soft* performs better than *hard*. However, in our current task, we used the hard voting method. If we had used soft voting, better results would have been achieved.

2.2.2 Implementation Step of Subtask 2 (Gender Bias Classification)

First, the *simpletransformers* Python library [10] that will be used below requires the data to be presented in a specific form. The following data samples adapts each sample split to contain only two columns: text and labels, where the text is an array equal in size to the number of labels [11]. Second, models' definition. In this model setup section, the different transformers that will be evaluated are gathered. For this purpose, the implementation mainly relies on the *simpletransformers* Python library, which allows to train and test transformers within few steps.

Third, training pre-trained models. Each of the aforementioned pre-trained models is trained separately with the entire training set. This training is directly performed in the previously defined dictionary for convenience. Fourth, ensembles' definition. The ensembles of transformers that can be defined with the previously specific trained models are created. A dictionary is created for convenience, univocally identifying each ensemble. Fifth, evaluation fine-tuning pre-trained models. Firstly, each transformer is individually evaluated using the validation split. Subsequently, the main evaluation metrics e.g. accuracy, F1-score, precision and recall are stored. Secondly, the predictions of each ensemble for the validation set instances are derived. After calculating their metrics, it is possible to determine which ensemble obtained the best Macro F1-score. This will be the final ensemble used for the test dataset. Regarding the ensembles' predictions, these are obtained through a hard voting system after computing the output that each of the ensemble's models produces for a given instance, the most-voted class turns out to be the ensemble result. The voting system can be non-weighted or weighted. In the latter, the prediction of each individual transformer is weighted according to their normalized F1-score, thus providing a greater importance to the best pre-trained model without disregarding the outputs of the other transformers. Sixth, the vote function determines the ensemble prediction based on the outcomes of its base transformers. Its arguments are: *predictions*, list of transformers' raw outputs. *weighted*, bool that determines if a weighted voting system must be used. *weights*, list of weights e.g. normalized weighted F1-scores. Seventh, selecting the best ensemble. Once the predicted labels for each validation instance are calculated for each ensemble, their metrics can be

computed. Given subtask 2 that it is a multilabel classification task, the best ensemble will be that with a maximum Macro F1-score. Eighth, predictions on official test set. Finally, the ensemble which obtained a higher Macro F1-score can be used to predict the label of each test instance. Further, these results will be used to portray some evaluation plots, including the confusion matrix for class AC, DI, and ANB, see Figures 3, 4 and 5 and the ROC curve, see Figure 6. The corresponding specific pre-trained model parameters are shown in Table 4 where the model learning name is *bert-cased*.

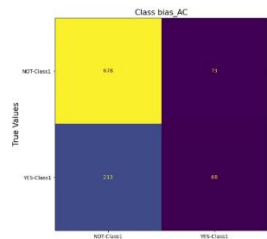


Fig. 3. A confusion matrix figure of Gender Stereotyped Activity and Career Choices (class AC).

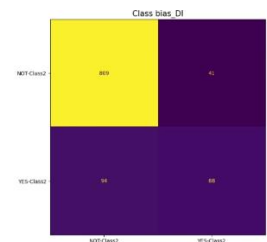


Fig. 4. A confusion matrix figure of Gender Stereotyped Descriptions and Inductions (class DI).

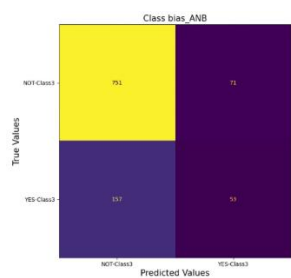


Fig. 5. A confusion matrix figure of Expressed Gender-stereotyped Attitudes, Norms and Beliefs (class ANB).

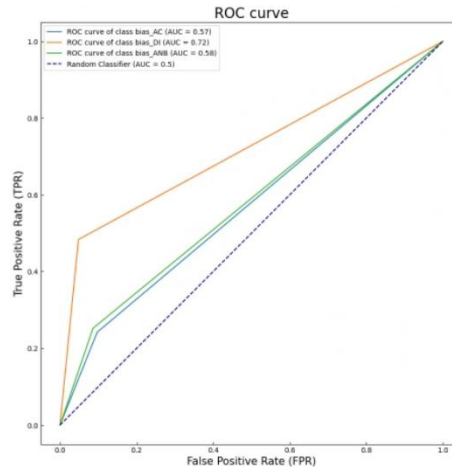


Fig. 6. A ROC curve figure of Gender Stereotyped Activity and Career Choices (class AC), Gender Stereotyped Descriptions and Inductions (class DI) and Expressed Gender-stereotyped Attitudes, Norms and Beliefs (class ANB).

Table 4. The hyper-parameters of *bert-cased* pre-trained model.

Parameter	Value
eval_batch_size	8
num_train_epochs	5
learning_rate	4e-05
optimizer	AdamW
manual_seed	42
use_early_stopping	True

3. Results

3.1. Results Analysis of Subtask 1 (Gender Bias Detection)

The following records the official results of Subtask 1 (Gender Bias Detection). The metrics recorded by the three different approaches in the evaluation task of the test set, see Table 5. Submission1, Submission2 and Submission3 respectively using the *bagging* model, the *bagging2* model and the *roberta-chinese-large* model on the test dataset. The specific situation of the pre-trained models are shown in Table 1. Perhaps because other pre-trained models except the *roberta-chinese-large* model did not perform well on this task, we found that fine-tuning the *roberta-chinese-large* pre-trained model alone achieved the best results instead.

Table 5. The official evaluation results of Subtask 1(Gender Bias Detection).

Submission	Model	Precision	Recall	F1 Score
1	Bagging	0.464	0.540	0.499
2	Bagging2	0.507	0.680	0.581
3	<i>roberta-chinese-large</i>	0.565	0.990	0.719

3.2. Results Analysis of Subtask 2 (Gender Bias Classification)

The following records the official results of Subtask 2 (Gender Bias Classification). The metrics recorded by the three different approaches in the evaluation task of the test set, see Table 6. Submission1, Submission2 and Submission3 respectively using the *distilbert-multi* model, the bagging model and the bagging2 model on the test dataset. The specific situation of the model is shown in Table 2. Perhaps because other pre-trained models except the *distilbert-multi* model did not perform well on this task, we found that fine-tuning the *distilbert-multi* model alone achieved the best results instead.

Table 6. The official evaluation results of Subtask 2 (Gender Bias Classification).

Submission	Model	Precision	Recall	F1 Score
1	<i>distilbert-multi</i>	0.510	0.590	0.547
2	Bagging	0.490	0.264	0.343
3	Bagging2	0.484	0.283	0.357

3.3. Extend Results Analysis of Subtask 2 (Gender Bias Classification)

The following records the unofficial results of Subtask 2 (Gender Bias Classification). The metrics recorded by the three different approaches in the evaluation task of the validation set, see Table 7. The NLPCC 2025 Shared Task 7-CORGI-PM 2025: Chinese Corpus for Gender Bias Detection, Classification and Mitigation Challenge official organizer replied: I'm sorry that I could not accept more submissions from your team since your scores have already been released. We will only release your scores as your final results to help you finish your technical report instead of modifying your method. So these results are only for reference because there are no official test result for them. Submission1, Submission2 and Submission3 respectively using the *bert-cased* model, the bagging model and the bagging2 model on the validation dataset. The specific situation of the pre-trained models are shown in Table 3. Perhaps because other pre-trained models except the *bert-cased* model did not perform well on this specific task, we found that fine-tuning the *bert-cased* model alone achieved the best results instead.

Table 7. The unofficial evaluation results of Subtask 2 (Gender Bias Classification).

Submission	Model	Accuracy	Precision (Macro)	Recall (Macro)	F1 Score (Macro)	F1 Score (Weighted)
1	<i>bert-cased</i>	0.65620	0.63062	0.52596	0.57355	0.55663
2	Bagging	0.53198	0.54080	0.21127	0.35044	0.33003
3	Bagging2	0.55620	0.53062	0.32596	0.40384	0.38663

4. Discussion

The ranking analysis that should be used for given references in the results section. Official test evaluation results content should be used for ranking, compared with the experimental results of other teams.

4.1. Ranking Analysis of Subtask 1 (Gender Bias Detection)

The following records the official results ranking of Subtask 1 (Gender Bias Detection). The metrics recorded by the best approach in the evaluation task of the test set, see Table 8. The ranking is based on the best F1 score value of the test dataset evaluation results submitted by each team.

Table 8. The official results ranking of Subtask 1 (Gender Bias Detection).

Ranking	Team Name	Precision	Recall	F1 Score
1	ZZU-nlp	0.798	0.910	0.850
2	ZZU-NLP-DS	0.604	0.960	0.741
3	Cloud Lab	0.566	0.990	0.720
4	wangkongqiang	0.565	0.990	0.719
5	YNU-HPCC	0.645	0.800	0.714
6	Prompt	0.552	1.000	0.712
7	IR901	0.557	0.980	0.710
8	xlarge	0.536	0.960	0.688
9	Hu	0.522	0.970	0.678
10	Team0071	0.496	0.640	0.559

4.2. Ranking Analysis of Subtask 2 (Gender Bias Classification)

The following records the official results ranking of Subtask 2 (Gender Bias Classification). The metrics recorded by the best approach in the evaluation task of the test set, see Table 9. The ranking is based on the best F1 score value of the test dataset evaluation results submitted by each team.

Table 9. The official results ranking of Subtask 2 (Gender Bias Classification).

Ranking	Team Name	Precision	Recall	F1 Score
1	ZZU-nlp	0.543	0.576	0.646
2	Team0071	0.511	0.591	0.548
3	wangkongqiang	0.510	0.590	0.547
4	Hu	0.447	0.658	0.531
5	YNU-HPCC	0.463	0.565	0.509
6	Prompt	0.496	0.513	0.505
7	Tsingyi	0.479	0.468	0.474
8	Cloud Lab	0.510	0.409	0.453
9	IR901	0.443	0.420	0.431
10	ZZUNLP-DS	0.466	0.361	0.407

5. Conclusion

For pre-trained models references, we prefer to the use of pre-trained models that have been fine-tuned on other relative corpus datasets and use the integrated model of common decision-making [12]. Although the integrated model of common decision-making is used, it does not show a significant advantage over individual excellent fine-tuned pre-trained models. However, the performance of the integrated model using common decision-making on the test set are also acceptable. In the following steps, we will provide a sample reference list with select the fine-tuned pre-trained models that perform well on these chinese corpus dataset. It is the future work goal of further improving performance and increase evaluation F1 scores.

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